

Fuse operations around data reuse

Mathematical stages at left; audited execution boundaries at right. Supported paths are shape / dtype / GPU specific.

OPERATOR	MATHEMATICAL STAGES · NOT A BASELINE LAUNCH COUNT	IMPLEMENTED GROUPING
TriMul Inference [C1, C3]	LN → input projections / sigmoid gates / mask → contraction → LN / projection / gate / residual	K1 LN + proj + gate + mask → cuBLAS contraction(s) → K3 LN + proj + gate + add A/B and contraction output still cross HBM; bidirectional inference uses two BMM calls.
Transition BF16 D128; n=4 [C4]	LN → expand A/B → SwiGLU → squeeze → residual	One fused forward kernel LN + expand + SwiGLU + squeeze + residual Training saves normalized input / statistics. Backward also has a separate partial reduction.
OPM MSA training [C5]	LN → two projections + mask → outer product → permute / mask-count normalization → output projection	Triton prologue LN + projections + mask → cuBLAS grouped product → CUDA epilogue layout + norm + projection Mask-count normalization is computed separately. Save O or recompute it in backward.
PWA MSA training [C6]	Pair: LN / projection / mask / softmax MSA: LN / value → weighted sum / gate / output / residual	Pair: LN + projection + softmax MSA: LN + value projection → Weighted sum + gate output + residual First two blocks are parallel input branches, not a serial chain. Training also prepares a dropout mask.
LNLinear CuTe SM90 path [C7]	LN statistics → normalized activation → linear projection	One main kernel GEMM + row statistics + affine correction epilogue Avoids materializing the full normalized activation; weight-side preparation is outside this box.
Triangle Attention Dispatch-dependent [C8, C9]	LN / QKV / bias → tiled attention → sigmoid gate → output projection	Projection path → Tiled attention → Gated output path Dashed boxes denote stages, not single launches. Projection and output fusion depend on dispatch.
LayerNorm RMSNorm Standalone primitives	Row statistics / reduction → normalization + optional affine	Standalone normalization row reduction + normalization / affine A primitive optimization is not evidence of cross-operator fusion. Backward may require partial reductions.

SOLID BORDER audited main kernel / library-call group DASHED BORDER execution stage AMBER ARROW retained global-memory boundary